Data Session 5 Key, part 2: The Money Nobody Controls

84-355 Democracy's Data | Session 5 — Independent expenditures | Instructor copy — do not distribute

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All code is given to students. Grade the interpretation.

Where this sits. The second half of Session 5, after the candidates' own FEC filings. **It replaced a planned lab on the Meta and Google ad libraries, which turned out to be closed** — Meta needs a token and identity verification; Google publishes only through BigQuery, in spend ranges rather than figures. The syllabus now spends a minute on that absence, and it is worth making: the most detailed record of political persuasion in America is not public.

This is the dirtiest dataset in the course, and it is the point.

Every number below was run against the data files.

The session in one sentence

The 16 rows in `data/ie_outliers.csv`, out of 73,449 in the source file, inflate the federal record of outside spending 10-fold and reverse its most important finding, and they are still there.

The findings

```r
c(rows_total     = ROWS,          # 73,449; see setup - the raw file is not committed
  outlier_rows   = nrow(o),
  raw_billions   = round(sum(s$dollars[s$version == "raw"]) / 1e9, 1),
  clean_billions = round(sum(s$dollars[s$version == "cleaned"]) / 1e9, 2),
  raw_pct_oppose   = s$pct[s$version == "raw"     & s$side == "oppose"],
  clean_pct_oppose = s$pct[s$version == "cleaned" & s$side == "oppose"])
```

```
#>       rows_total     outlier_rows     raw_billions   clean_billions
#>         73449.00            16.00            49.70             4.75
#>   raw_pct_oppose clean_pct_oppose
#>             5.50            57.60
```

- **$49.7bn raw, $4.75bn real.** 16 rows carry **90.4%** of the raw total.
- **Support/oppose flips: 94.5/5.5 raw → 42.4/57.6 cleaned.**
- **Senate races are 75.1% negative, House 68.2%, President 43.5%.**
- Bob Casey: **87.6%** of the outside money around him was spent against him.

The three things to get said

1. There are two different problems here and students must separate them. The junk filings (*THE COURT OF DIVINE JUSTICE*, *$255M for Walter White*) are **people lying to the government**. The 9-fold Food & Water duplicate is **the government's own export repeating a row**. One is fraud, one is plumbing, and they need different fixes. **Do not let the room lump them together as "bad data."**

2. The flip is the lesson, not the total. Anyone can see that $49.7bn is too big. The dangerous error is the one that looks fine: 94.5% supportive is a perfectly plausible number that a student could put in a paper. **Wrongness that announces itself is the safe kind.** Ask what else they have read that had a quiet version of this problem.

3. "The FEC publishes what is filed." It is a disclosure agency, not an auditor. Ask what checking would cost and who would do it — and note that the alternative, an agency that decides which filings are real, has obvious problems of its own. **There is no free version of this.**

The line for the board: official does not mean checked.

Option A — Argue with the threshold

Verified output

cut at	rows removed	total	support / oppose
none (raw)	0	$49.7bn	94.5 / 5.5
$1M	897	$2.39bn	47.7 / 52.3
$5M	102	$3.92bn	39.8 / 60.2
$10M	42	$4.34bn	40.9 / 59.1
$50M	**16**	**$4.75bn**	**42.4 / 57.6**
$100M	**16**	**$4.75bn**	**42.4 / 57.6**
$500M	6	$6.03bn	54.7 / 45.3

The three middle rows are the only figures in this key that are **not** re-derived from `data/`. `ie_outliers.csv` holds the rows above the $50M cut, so every threshold from $50M up recomputes here; the $1M, $5M and $10M rows were recorded from the uncommitted 19 MB raw file when `data/build-ie-data.R` ran.

3/4: Runs two or three thresholds and reports that the answer moves.

4/4: Sees that **the flip survives everything from $1M to $100M** — every threshold in that range gives roughly 40/60 — so the finding is robust across two orders of magnitude. That is a much stronger claim than "it worked at the number I picked."

4/4, best: Works out **why $500M breaks it.** Raising the cut that far lets 10 rows back in, every one of them on the *supporting* side: the 9 identical Food & Water rows, $1.03bn of duplicated money, together with the $255M Walter White filing. So the sensitivity is not vagueness in the data; it is two specific artifacts re-entering — one a duplicate, one a fabrication — and a student who diagnoses either has done real work.

Also worth noticing: **$50M and $100M give identical answers**, because no row sits between them — the smallest row removed at either cut is $114M. A threshold with slack around it is more defensible than one tuned to the data.

Option B — A better rule

No single answer; grade the design.

3/4: Proposes a plausible rule — flag committees that filed once, or amounts above the cycle's known total.

4/4: Says what the rule would wrongly remove. A single-filing rule kills every small genuine committee that spent once. A "purpose text looks fake" rule requires somebody to judge text, which is the discretion problem from Session 11 in a new place.

4/4, best: Notices that the honest fix is not a rule at all but **publication of the 16 rows alongside the total**, so a reader can decide. That is what this lab does, and a student who arrives at it independently should be told so.

Option C — Negative by office

Verified output

```
cd$off <- c(P = "President", S = "Senate", H = "House")[cd$office]
aggregate(cbind(supporting, opposing) ~ off, data = cd, sum)
```

```
#>         off supporting  opposing
#> 1     House   11990012  25763543
#> 2 President 1184801494 913938021
#> 3    Senate  315312780 949275808
```

Across the committed file: **Senate 75.1% against, House 68.2%, President 43.5%.**

(An earlier version of this key reported 71.5 / 69.1 / 42.6. Those were computed on the full cleaned FEC extract, which is not what ships in `data/` — and 42.6 was Harris's own `pct_against`, not the presidential aggregate. The figures above re-derive from `ie_candidates.csv` and are what the chunk prints. Every number in a key should reproduce from the committed data; these now do.)

3/4: Reports the ranking.

4/4: Explains the presidential exception. Both 2024 presidential candidates had very large *supportive* super PACs; down-ballot, outside groups mostly attack, because attacking an unknown candidate is cheaper than introducing your own. **Name recognition is the mechanism.**

4/4, best: Notes the strategic asymmetry — a negative ad about a little-known Senate challenger can define them before they define themselves, which is not available against a sitting president everybody already has an opinion about. **The spending pattern is a function of how much the electorate already knows.**

Grading

Four-point scale from the syllabus.

- **Never penalise code.** It is your code.
- **The move that earns a 4** is separating fraud from plumbing, and noticing that the plausible error is more dangerous than the absurd one.
- **A 2** is "you can't trust government data." That is both lazy and wrong — the data is fine once you look at it, and looking took 16 rows.
- **Reward anyone who asks what the FEC is for.** Disclosure and verification are different jobs and Congress only funded one of them.

Anticipated questions

- *"Is filing a fake report a crime?"* — Knowingly false FEC filings can be. Whether these were prosecuted is not in this file; do not speculate.
- *"How did nobody catch this?"* — Somebody probably did. The bulk file is the raw record; the FEC's own summary pages apply filters this file does not. **That is worth saying: the same agency publishes both a clean and a dirty version, and the dirty one is the one you download.**
- *"Who is Nader Akhlaghy?"* — A minor candidate named in the junk filings. We do not know why, and the file cannot say.
- *"Why is 'independent' spending independent?"* — Because *Buckley* and *Citizens United* permit unlimited spending only if uncoordinated. It is a legal category, and whether it describes reality is what enforcement fights are about. This connects straight to Tuesday's Ch. 4 debate.

Connections

- **Session 5 part 1 (FEC candidate filings)** — same agency, same cycle, and a file that is clean. **Contrast them explicitly; it shows the problem is the form, not the FEC.**
- **Session 11 (EAVS)** — the other self-reported dataset, and the other place where the reporting entity is the entity being measured.
- **Session 1 (returns)** — the first "two official sources disagree" moment.
- **Session 14 (RPV)** — where a defensible-looking methodological choice changes a legal conclusion. Same shape as the threshold here.